

HUB REMOVAL IN RANDOMLY ORIENTED DIVISOR GRAPHS: THE GIANT STRONG COMPONENT IS THE SMOOTH FIBRE

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ABSTRACT. Let $\mathcal{D}_\rho(N)$ be the randomly oriented divisor graph of Kim and Phillips: the divisor graph on $\{1, \dots, N\}$, with each edge oriented from multiple to divisor and then independently reversed with probability ρ . Let Φ_K be the size of the largest strongly connected component after the vertices $1, \dots, K$ are deleted. We prove that for every fixed $\rho \in (0, 1)$,

$$0 \leq \Psi\left(N, \frac{N}{K+1}\right) - \mathbb{E}[\Phi_K] \leq \frac{30N}{\log \log N} \quad (0 \leq K < N, N \geq N_0(\rho)),$$

where $\Psi(x, y)$ is the number of y -smooth integers up to x . Consequently, if $K = N^{\theta+o(1)}$ with $0 \leq \theta \leq 1$, then $\mathbb{E}[\Phi_K]/N \rightarrow \rho_{\text{Dick}}(1/(1-\theta))$, where ρ_{Dick} is Dickman's function and $\rho_{\text{Dick}}(\infty) = 0$. The same limit holds if the K vertices of largest degree are deleted instead, whether degrees are computed once or recomputed after each deletion. This answers Question 20 of Kim and Phillips. Moreover, with high probability the largest strong component consists of all but $o(N) + O(K)$ of the $N/(K+1)$ -smooth vertices, and the second largest strong component has expected size at most $K + o(N)$.

The upper bound comes from the decomposition of the divisor graph into fibres of the rough part. The lower bound rests on one observation. Every divisibility edge $\{x, x'\}$ with $x' \leq N^{1-2\delta}$ is shadowed by at least $N^{2\delta} - 2$ independent directed 2-paths through common multiples. So with high probability the strong components are at least as coarse as the connected components of a deterministic divisor graph on $(K, N^{1-2\delta}]$, and a doubling argument shows that the smooth part of that graph is connected. The proof is elementary: it uses only the theorems of Chebyshev and Mertens, a second-moment bound for the number of small prime factors, and, to identify the limit, Dickman's asymptotic for Ψ .

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1. INTRODUCTION

1.1. The model and the question. For $N \geq 2$ the *divisor graph* G_N has vertex set $[N] = \{1, \dots, N\}$ and an edge $\{a, b\}$ whenever $a \neq b$ and $a \mid b$ or $b \mid a$. Kim and Phillips [9] introduced the *randomly oriented divisor graph* $\mathcal{D}_\rho(N)$. Each edge $\{d, m\}$ with $d \mid m$, $d < m$, is first oriented

$m \rightarrow d$ (multiple to divisor) and then independently reversed with probability $\rho \in [0, 1]$. Thus every edge independently points $m \rightarrow d$ with probability $1 - \rho$ and $d \rightarrow m$ with probability ρ . Write $\#\Phi(D)$ for the number of vertices of a largest strongly connected component (SCC) of a directed graph D . The main theorem of [9] states that $\mathbb{E}[\#\Phi(\mathcal{D}_\rho(N))]/N \rightarrow 1$ for every fixed $\rho \in (0, 1)$. The authors ask:

Question 1.1 ([9, Question 20]). What can be said about the expected size of the largest strongly connected component as we remove the vertices of highest degrees, one by one. In other words, how will the expected size of the largest strongly connected component change as we remove the vertices labeled 1, 2, 3, 4, and so on?

For $0 \leq K < N$ let $V_{N,K} = \{K+1, \dots, N\}$, let $G_{N,K} = G_N[V_{N,K}]$ be the induced subgraph, and let

$$\mathcal{D}_\rho(N, K) := \mathcal{D}_\rho(N)[V_{N,K}]$$

be the random directed graph obtained from $\mathcal{D}_\rho(N)$ by deleting the vertices $1, \dots, K$. Its edges are exactly the edges of $G_{N,K}$, oriented independently as above. Put

$$\Phi_K := \#\Phi(\mathcal{D}_\rho(N, K)).$$

Throughout, $\rho \in (0, 1)$ is the reversal probability, and we write

$$q := \rho(1 - \rho) \in (0, \tfrac{1}{4}], \quad \lambda := \max(\rho, 1 - \rho) \in [\tfrac{1}{2}, 1).$$

1.2. Notation. p always denotes a prime. $P^+(n)$ is the largest prime factor of n , with $P^+(1) = 1$. For real $x \geq 0$ and $y \geq 1$,

$$\Psi(x, y) := \#\{1 \leq n \leq x : P^+(n) \leq y\}.$$

For $z \geq 2$, $\omega_z(n) := \#\{p \leq z : p \mid n\}$. We write ρ_{Dick} for Dickman's function (§2); the letter ρ alone always means the reversal probability. For $0 \leq K < N$ put

$$B = B(N, K) := \left\lfloor \frac{N}{K+1} \right\rfloor, \quad F_1 = F_1(N, K) := \{m \in V_{N,K} : P^+(m) \leq B\}.$$

Note that $P^+(m) \leq N/(K+1)$ if and only if $P^+(m) \leq B$, so $\Psi(N, N/(K+1)) = \Psi(N, B)$. We call F_1 the *smooth fibre*. Logarithms are natural. “With high probability” (w.h.p.) means with probability $1 - o(1)$ as $N \rightarrow \infty$.

1.3. Results. Each statement carries a status tag. [PROVEN] means a complete proof is given in this paper.

Theorem 1.2 (Main theorem). [PROVEN] *Fix $\rho \in (0, 1)$. For all $N \geq 2$ and $0 \leq K < N$ we have $\Phi_K \leq \Psi(N, N/(K+1))$ deterministically, and*

$$\lim_{N \rightarrow \infty} \max_{0 \leq K < N} \frac{1}{N} \left(\Psi\left(N, \frac{N}{K+1}\right) - \mathbb{E}[\Phi_K] \right) = 0.$$

Theorem 1.3 (Rate). [PROVEN] *For every $\rho \in (0, 1)$ there is $N_0(\rho)$ such that for all $N \geq N_0(\rho)$ and all $0 \leq K < N$,*

$$0 \leq \Psi\left(N, \frac{N}{K+1}\right) - \mathbb{E}[\Phi_K] \leq \frac{30N}{\log \log N}.$$

For $K \leq \sqrt{N} - 1$ this becomes fully explicit (Remark 5.1):

$$(1) \quad 1 - \frac{\mathbb{E}[\Phi_K]}{N} = \log \frac{\log N}{\log B} + O_\rho\left(\frac{1}{\log \log N}\right), \quad B = \left\lfloor \frac{N}{K+1} \right\rfloor.$$

So the expected fraction of vertices lost from the giant after removing the hubs $1, \dots, K$ is asymptotically $\log \frac{\log N}{\log(N/K)}$, which is about $\frac{\log K}{\log N}$ when $K = N^{o(1)}$.

Corollary 1.4 (Limit profile). [PROVEN] Fix $\rho \in (0, 1)$, and let $K = K(N) \in [0, N]$ satisfy $\log(K+1)/\log N \rightarrow \theta \in [0, 1]$. Then

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[\Phi_K]}{N} = \rho_{\text{Dick}}\left(\frac{1}{1-\theta}\right),$$

with the convention $\rho_{\text{Dick}}(\infty) = 0$. The profile $\theta \mapsto \rho_{\text{Dick}}(1/(1-\theta))$ is continuous and strictly decreasing on $[0, 1]$. It equals $1 - \log \frac{1}{1-\theta}$ for $0 \leq \theta \leq \frac{1}{2}$.

Corollary 1.5 (Structure of the giant). [PROVEN] Fix $\rho \in (0, 1)$. Let Φ_K^* be a largest SCC of $\mathcal{D}_\rho(N, K)$ (chosen by any rule), and let $\Phi_K^{(2)}$ be the size of a largest SCC other than Φ_K^* (0 if there is none). For $N \geq N_0(\rho)$ and all $0 \leq K < N$:

- (a) $\mathbb{E}|\Phi_K - \Psi(N, \frac{N}{K+1})| \leq 30N/\log \log N$. In particular $\Phi_K/N - \Psi(N, \frac{N}{K+1})/N \rightarrow 0$ in probability, uniformly in K .
- (b) $\mathbb{E}|F_1 \triangle \Phi_K^*| \leq 2K + 30N/\log \log N$.
- (c) $\mathbb{E}[\Phi_K^{(2)}] \leq K + 30N/\log \log N$.

So when $K = o(N)$, w.h.p. the largest SCC is, up to $o(N)$ vertices, the set of $N/(K+1)$ -smooth integers in $(K, N]$, and it is unique in the sense that all other SCCs have $o(N)$ vertices. The term K in (c) cannot be dropped in general; see Remark 5.2.

Corollary 1.6 (Removing the vertices of highest degree). [PROVEN] Fix $\rho \in (0, 1)$ and let $\log(K+1)/\log N \rightarrow \theta \in [0, 1]$. Let $T = T(N) \subset [N]$ with $|T| = K$ be chosen from the undirected graph G_N alone (and so independently of the orientation), in either of the following ways:

- (i) static attack: $\min_{t \in T} \deg_{G_N}(t) \geq \max_{v \notin T} \deg_{G_N}(v)$;
- (ii) adaptive attack: $T = \{u_1, \dots, u_K\}$, where for each j the vertex u_{j+1} has maximal degree in $G_N - \{u_1, \dots, u_j\}$.

Ties may be broken arbitrarily. Then $\mathbb{E}[\#\Phi(\mathcal{D}_\rho(N) - T)]/N \rightarrow \rho_{\text{Dick}}(1/(1-\theta))$. Moreover, for fixed $\epsilon > 0$ and $N \geq N_1(\epsilon)$, if $K \leq N^{1/2-\epsilon}$ then the static set T is exactly $\{1, \dots, K\}$.

Since the total degree of a vertex in any orientation equals its degree in G_N , (i) and (ii) cover attacks by total degree in the directed graph. Attacks by in- or out-degree depend on the orientation and are not covered; see §7.

θ	0.1	0.2	1/3	0.4	1/2	0.6
$u = \frac{1}{1-\theta}$	1.111	1.25	1.5	1.667	2	2.5
$\rho_{\text{Dick}}(u)$	0.8946	0.7769	0.5945	0.4892	0.3069	0.1303
θ	2/3	3/4	0.8	0.9	0.95	1
$u = \frac{1}{1-\theta}$	3	4	5	10	20	∞
$\rho_{\text{Dick}}(u)$	0.04861	0.004911	$3.547 \cdot 10^{-4}$	$2.770 \cdot 10^{-11}$	$2.46 \cdot 10^{-29}$	0

TABLE 1. The limit profile $\lim \mathbb{E}[\Phi_K]/N$ for $K = N^{\theta+o(1)}$. For $u \leq 2$, $\rho_{\text{Dick}}(u) = 1 - \log u$. The other values were computed numerically from $u\rho_{\text{Dick}}(u) = \int_{u-1}^u \rho_{\text{Dick}}(t) dt$ and agree with the standard tables.

1.4. Discussion. Answer to Question 1.1. Deleting any $K = N^{o(1)}$ hubs costs nothing asymptotically; for instance $\mathbb{E}[\Phi_K]/N \rightarrow 1$ for every fixed K . Deleting $K = N^\theta$ hubs, a vanishing fraction of the vertices, leaves a giant of relative size $\rho_{\text{Dick}}(1/(1-\theta))$. For example, deleting the $\sqrt{N}$ hubs leaves $1 - \log 2 \approx 30.7\%$. Deleting $N^{1-o(1)}$ hubs destroys the giant. The vertices $1, \dots, K$ carry a θ -fraction of the $N \log N + O(N)$ edges (Remark 7.1), so hub removal is far more damaging than the loss of edges alone suggests. This is the fragility to targeted attacks familiar from scale-free networks [1, 2].

In the divisor graph, however, the surviving giant has an explicit arithmetic description: the smooth fibre. At finite N the convergence is slow (compare Theorem 1.3). For instance, for $K = 1$ every prime $p \in (N/3, N/2]$ lies in F_1 but has degree 1 in $G_{N,1}$ (its only neighbour is $2p$), so it is never in a strong component of size ≥ 2 . Hence $\Psi(N, B) - \mathbb{E}[\Phi_1] \geq \pi(N/2) - \pi(N/3) \sim N/(6 \log N)$, a loss that vanishes asymptotically but is visible at moderate N (see (O1) in §7).

What is new. To our knowledge Theorems 1.2 and 1.3 and Corollaries 1.4–1.6 are new. The fibre decomposition behind the upper bound (Lemma 3.1) is a simple and essentially known observation; a closely related statement appears in McNew’s work on the divisor graph [10]. The inputs from analytic number theory (Mertens, Chebyshev, Dickman, a Turán-type variance bound, and the divisor bound) are classical. The lower-bound mechanism (Lemmas 4.1 and 4.2) is, as far as we know, new. It differs from the triangle argument of [9], which uses paths through the vertex 1, and it does not need a strongly connected “hub clique”. Such a clique, for instance the interval $(K, N^{1/2-\delta}]$, whose elements pairwise have many common multiples, can capture only the integers with a divisor in the clique. For $\theta > 1/3$ that should miss a positive proportion of F_1 ; see the sketch in Remark 7.2.

1.5. Idea of the proof. *Upper bound.* If $d \mid m$ with $K < d < m \leq N$, then $m/d \leq B$. Hence the B -rough part of a vertex is constant on connected components, and every SCC lies in a fibre $\{v : v = ns, P^+(s) \leq B\}$, which has at most $\Psi(N, B)$ elements.

Lower bound. Fix a small $\delta > 0$ and put $M = N^{1-2\delta}$.

- (1) *Robust edges.* If $K < x < x' \leq M$ and $x \mid x'$, then each of the $\geq N^{2\delta} - 2$ multiples y of x' in $(x', N]$ gives a directed path $x \rightarrow y \rightarrow x'$ with probability q , independently over y ; likewise for $x' \rightarrow y \rightarrow x$. A union bound shows that w.h.p. every edge of the *deterministic* graph $G_{M,K}$ joins two vertices of the same SCC. So every connected component of $G_{M,K}$ lies inside one SCC.
- (2) *Doubling.* All $x \in (K, M]$ with $P^+(x) \leq M/(2(K+1))$ lie in one component of $G_{M,K}$. Multiply by 2 until just below M , divide by an odd prime, and repeat; this ends at the power of 2 in $(M/2, M]$. Call this set S_M .
- (3) *Attachment.* A vertex $m \in (M, N]$ joins the SCC containing S_M as soon as, among the edges from m to its divisors in S_M , at least one points each way. This has probability $1 - \rho^s - (1 - \rho)^s$, where s is the number of such divisors. An elementary counting argument shows that all but $o(N)$ elements of F_1 have $s \geq s_0$ for any fixed s_0 . It uses a reserve of s_0 small prime factors of m (supplied by Turán’s variance bound), a deterministic case analysis, and Mertens’ theorem.

Organisation. §2 collects the arithmetic inputs. §3 proves the upper bound, §4 the general lower bound (Proposition 4.6), and §5 the theorems and corollaries. §6 treats degree-based removal, and §7 contains remarks and open problems. Appendix A proves the sub-sum lemma from the preliminary notes, which is no longer needed. Appendix B lists every claim with its status.

2. ARITHMETIC PRELIMINARIES

We use the following classical facts. The cited ones are stated with their hypotheses.

Lemma 2.1 (Mertens [11], Chebyshev; see [6, Ch. XXII], [12, Part I, Ch. 1]). [PROVEN, standard; cited] *There are absolute constants $c_0, C_1 \geq 1$ such that:*

- (a) $\sum_{p \leq y} \frac{\log p}{p-1} \leq \log y + c_0$ for all $y \geq 2$;
- (b) $|\sum_{y < p \leq w} \frac{1}{p} - \log \frac{\log w}{\log y}| \leq \frac{C_1}{\log y}$ for all $2 \leq y \leq w$;
- (c) $\sum_{p \leq z} \frac{1}{p} \geq \log \log z - C_1$ for all $z \geq 2$;
- (d) $\pi(x) \leq C_1 x / \log x$ for all $x \geq 2$.

Source. Mertens' first theorem, $\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1)$ for $x \geq 2$, together with $\sum_p \frac{\log p}{p(p-1)} < \infty$, gives (a). Mertens' second theorem, $\sum_{p \leq x} \frac{1}{p} = \log \log x + b_1 + O(\frac{1}{\log x})$ for $x \geq 2$ with an absolute constant b_1 , gives (b) by subtraction and (c) directly. Part (d) is Chebyshev's upper bound [6, Thm. 7]. $\square$

Lemma 2.2 (Dickman's function [4, 3, 7]; see [12, Part III, Ch. 5], [5, 8]). [PROVEN, standard; cited] *Let $\rho_{\text{Dick}} : [0, \infty) \rightarrow (0, 1]$ be the unique continuous function with $\rho_{\text{Dick}}(u) = 1$ for $0 \leq u \leq 1$ and $u \rho'_{\text{Dick}}(u) = -\rho_{\text{Dick}}(u-1)$ for $u > 1$. Then ρ_{Dick} is positive, $\rho_{\text{Dick}}(u) = 1 - \log u$ for $1 \leq u \leq 2$, and ρ_{Dick} is strictly decreasing on $[1, \infty)$. For every fixed $U \geq 1$,*

$$\sup_{1 \leq u \leq U} \left| \frac{\Psi(x, x^{1/u})}{x} - \rho_{\text{Dick}}(u) \right| \rightarrow 0 \quad (x \rightarrow \infty).$$

Source. The pointwise statement is due to Dickman [4]; de Bruijn [3] gave an error term. Uniformity for u in compact sets follows, for example, from Hildebrand's theorem [7]: $\Psi(x, y) = x \rho_{\text{Dick}}(u) (1 + O_\varepsilon(\log(u+1)/\log y))$ uniformly for $x \geq 3$ and $\exp((\log \log x)^{5/3+\varepsilon}) \leq y \leq x$. For $u \leq U$ we have $y = x^{1/u} \geq x^{1/U}$, which lies in this range for large x . Positivity and $\rho_{\text{Dick}}(u) = 1 - \log u$ on $[1, 2]$ are standard. Strict decrease follows from $\rho'_{\text{Dick}}(u) = -\rho_{\text{Dick}}(u-1)/u < 0$. $\square$

Lemma 2.3 (Elementary smooth-number bound). [PROVEN, standard] *For $x \geq 2$ and $2 \leq y \leq x$,*

$$\Psi(x, y) \leq \sqrt{x} + \frac{2x(\log y + c_0)}{\log x}.$$

In particular $\rho_{\text{Dick}}(u) \leq 2/u$ for all $u \geq 1$, and $\rho_{\text{Dick}}(u) \rightarrow 0$ as $u \rightarrow \infty$.

Proof. For $n \geq 1$ we have $\log n = \sum_{p^k | n, k \geq 1} \log p$. Hence

$$\begin{aligned} \sum_{\substack{n \leq x \\ P^+(n) \leq y}} \log n &= \sum_{\substack{p \leq y \\ k \geq 1}} \log p \cdot \#\{n \leq x : P^+(n) \leq y, p^k | n\} \\ &\leq x \sum_{p \leq y} \log p \sum_{k \geq 1} p^{-k} = x \sum_{p \leq y} \frac{\log p}{p-1} \leq x(\log y + c_0) \end{aligned}$$

by Lemma 2.1(a). On the other hand, at least $\Psi(x, y) - \sqrt{x}$ of the n counted satisfy $n > \sqrt{x}$, and each of them has $\log n > \frac{1}{2} \log x$. Therefore $(\Psi(x, y) - \sqrt{x}) \cdot \frac{1}{2} \log x \leq x(\log y + c_0)$.

For the last assertion, put $y = x^{1/u}$ and let $x \rightarrow \infty$ in Lemma 2.2. This gives $\rho_{\text{Dick}}(u) \leq \lim_{x \rightarrow \infty} (x^{-1/2} + 2(\frac{1}{u} \log x + c_0)/\log x) = 2/u$. $\square$

Uses: Lemma 2.1(a); for the consequence, Lemma 2.2.

Lemma 2.4 (Variance of ω_z ; the special case of Turán–Kubilius we need). [PROVEN, standard] *Let $N \geq 1$ be an integer, $2 \leq z \leq N$, and $L := \sum_{p \leq z} \frac{1}{p}$. Then*

$$\sum_{m \leq N} (\omega_z(m) - L)^2 \leq 3NL.$$

Consequently, for every real $0 \leq s \leq L/2$,

$$\#\{m \leq N : \omega_z(m) < s\} \leq \frac{12N}{L}.$$

Proof. Write $f = \omega_z$. We have $S_1 := \sum_{m \leq N} f(m) = \sum_{p \leq z} \lfloor N/p \rfloor \geq NL - \pi(z)$. Also

$$S_2 := \sum_{m \leq N} f(m)^2 = \sum_{p \leq z} \lfloor N/p \rfloor + \sum_{\substack{p, p' \leq z \\ p \neq p'}} \lfloor N/(pp') \rfloor \leq NL + NL^2,$$

because m is divisible by the distinct primes p, p' if and only if $pp' \mid m$. Hence

$$\sum_{m \leq N} (f(m) - L)^2 = S_2 - 2LS_1 + NL^2 \leq NL + NL^2 - 2L(NL - \pi(z)) + NL^2 = NL + 2L\pi(z) \leq 3NL,$$

using $\pi(z) \leq z \leq N$. If $f(m) < s \leq L/2$, then $|f(m) - L| > L/2$. By Chebyshev's inequality the number of such m is at most $3NL/(L/2)^2 = 12N/L$. $\square$

Uses: nothing beyond counting.

Remark 2.5 (The Turán–Kubilius inequality). Lemma 2.4 is the special case $f = \omega_z$ of the Turán–Kubilius inequality [13]; see [12, Part III, Ch. 3]. That inequality states the following. There is an absolute constant C such that for every complex additive function f and every $x \geq 2$,

$$\sum_{n \leq x} |f(n) - A(x)|^2 \leq C x B(x)^2,$$

where $A(x) = \sum_{p^\nu \leq x} f(p^\nu) p^{-\nu} (1 - p^{-1})$ and $B(x)^2 = \sum_{p^\nu \leq x} |f(p^\nu)|^2 p^{-\nu}$. We use only the self-contained special case proved above.

Lemma 2.6 (Divisor bound [6, Thm. 315]). [PROVEN, standard] *For every $\varepsilon > 0$ there is C_ε with $\tau(n) \leq C_\varepsilon n^\varepsilon$ for all $n \geq 1$.*

Proof. We have $\tau(n)/n^\varepsilon = \prod_{p^a \parallel n} (a+1)p^{-a\varepsilon}$. If $p \geq 2^{1/\varepsilon}$, then $p^{a\varepsilon} \geq 2^a \geq a+1$, so the factor is ≤ 1 . If $p < 2^{1/\varepsilon}$, the factor is at most $c_\varepsilon := \sup_{a \geq 0} (a+1)2^{-a\varepsilon} < \infty$. Hence $\tau(n) \leq c_\varepsilon^{2^{1/\varepsilon}} n^\varepsilon$. $\square$

3. THE UPPER BOUND: FIBRES OF THE ROUGH PART

For $n \geq 1$ let $R_B(n) := \prod_{p^v \parallel n, p > B} p^v$ be the B -rough part of n .

Lemma 3.1 (Fibres). [PROVEN; essentially known, cf. [10]] *Let $0 \leq K < N$ and $B = \lfloor N/(K+1) \rfloor$.*

- (a) *If $\{d, m\}$ is an edge of $G_{N,K}$ with $d < m$, then $m/d \leq B$, and hence $R_B(d) = R_B(m)$.*
- (b) *R_B is constant on every connected component of $G_{N,K}$.*
- (c) *For $n \geq 1$ let $V_n := \{v \in V_{N,K} : R_B(v) = n\}$. Then $|V_n| \leq \Psi(N/n, B) \leq \Psi(N, B)$. Moreover $V_1 = F_1$, $|F_1| = \Psi(N, B) - \Psi(K, B)$, and $|V_n| \leq K$ for $n > 1$.*

Proof. (a) Since $d \geq K+1$, we have $m/d \leq N/(K+1)$. As m/d is an integer, $m/d \leq B$. So every prime factor of m/d is at most B , and $v_p(m) = v_p(d)$ for every $p > B$.

(b) This follows from (a) by induction along paths.

(c) Each $v \in V_n$ factors uniquely as $v = ns$ with $s = v/n$ being B -smooth. Since $s \leq N/n$, the map $v \mapsto s$ injects V_n into $\{s \leq N/n : P^+(s) \leq B\}$. Clearly $V_1 = \{v \in (K, N] : P^+(v) \leq B\} = F_1$, and $|F_1| = \Psi(N, B) - \Psi(K, B)$. If $n > 1$, then n has a prime factor $> B$, so $n \geq B+1 > N/(K+1)$. Hence $|V_n| \leq N/n < K+1$. $\square$

Uses: nothing.

Proposition 3.2 (Upper bound). [PROVEN] *For every orientation of $G_{N,K}$, every SCC is contained in a single fibre V_n . Hence*

$$\Phi_K \leq \Psi(N, B) \quad \text{and} \quad \Phi_K \leq \max(|F_1|, K).$$

Proof. An SCC is connected in the underlying undirected graph, so it lies in a connected component of $G_{N,K}$, and hence in a fibre by Lemma 3.1(b). Now apply Lemma 3.1(c). $\square$

Uses: Lemma 3.1.

Corollary 3.3. [PROVEN] *If $\log(K+1)/\log N \rightarrow \theta \in [0, 1)$, then $\limsup_N \mathbb{E}[\Phi_K]/N \leq \rho_{\text{Dick}}(1/(1-\theta))$.*

Proof. This follows from Proposition 3.2 and the computation of $\lim \Psi(N, B)/N$ in the proof of Corollary 1.4 (§5), which uses only Lemmas 2.2 and 2.3. $\square$

Remark 3.4 (Equivalent forms). Suppose $K = o(N)$. Since $0 \leq \Psi(N, B) - |F_1| \leq K$, Proposition 3.2 shows that $\mathbb{E}[\Phi_K] = \Psi(N, B) + o(N)$ is equivalent to $\mathbb{E}[|F_1| - \Phi_K] = o(N)$. It is also equivalent to the following: w.h.p. the largest SCC has at least $|F_1| - o(N)$ vertices. (Use Markov's inequality for the nonnegative variable $\Psi(N, B) - \Phi_K$.) Theorem 1.2 establishes all of these forms.

4. THE LOWER BOUND

Throughout this section $N \geq 2$, $0 \leq K < N$ and $\delta \in (0, \frac{1}{8}]$ are fixed, and

$$M := N^{1-2\delta} \quad (\text{a real number}), \quad Y := \frac{M}{2(K+1)}, \quad S_M := \{x \in \mathbb{Z} \cap (K, M] : P^+(x) \leq Y\}.$$

We write $G_{M,K}$ for the induced subgraph of G_N on $\mathbb{Z} \cap (K, M]$. This is a subgraph of $G_{N,K}$, and it is not random.

4.1. Robust edges.

Lemma 4.1 (Robust edges). [PROVEN; new] *Let $\mathcal{R}$ be the event that for all integers x, x' with $K < x < x' \leq M$ and $x \mid x'$ there are vertices $y, y' \in V_{N,K}$, both multiples of x' , such that $x \rightarrow y \rightarrow x'$ and $x' \rightarrow y' \rightarrow x$ in $\mathcal{D}_\rho(N, K)$. Then*

$$\mathbb{P}(\mathcal{R}^c) \leq M^2(1-q)^{N^{2\delta}-2} \leq N^2 \exp(-q(N^{2\delta}-2)).$$

On $\mathcal{R}$, the two endpoints of every edge of $G_{M,K}$ lie in the same SCC of $\mathcal{D}_\rho(N, K)$. Consequently every connected component of $G_{M,K}$ lies in a single SCC.

Proof. Fix x, x' as in the statement and let $\mathcal{Y} := \{y \in \mathbb{Z} : x' < y \leq N, x' \mid y\}$. Then

$$|\mathcal{Y}| = \lfloor N/x' \rfloor - 1 \geq N/M - 2 = N^{2\delta} - 2.$$

For $y \in \mathcal{Y}$ we have $K < x < x' < y \leq N$ and $x \mid x' \mid y$. So $\{x, y\}$ and $\{x', y\}$ are edges of $G_{N,K}$, and their default orientations are $y \rightarrow x$ and $y \rightarrow x'$. Let E_y be the event $\{x \rightarrow y \text{ and } y \rightarrow x'\}$, which says that $\{x, y\}$ is reversed and $\{x', y\}$ is not. Then $\mathbb{P}(E_y) = \rho(1-\rho) = q$. For distinct y the pairs of edges involved are disjoint, so the events $E_y, y \in \mathcal{Y}$, are independent. Hence

$$\mathbb{P}\left(\bigcap_{y \in \mathcal{Y}} E_y^c\right) = (1-q)^{|\mathcal{Y}|} \leq (1-q)^{N^{2\delta}-2}.$$

The same bound holds for the events $E'_y = \{x' \rightarrow y \text{ and } y \rightarrow x\}$, which say that $\{x', y\}$ is reversed and $\{x, y\}$ is not. There are fewer than $M^2/2$ pairs (x, x') and two directions, so the union bound gives $\mathbb{P}(\mathcal{R}^c) \leq M^2(1-q)^{N^{2\delta}-2}$. The second inequality uses $M \leq N$ and $1-q \leq e^{-q}$.

On $\mathcal{R}$, if $\{x, x'\}$ is an edge of $G_{M,K}$ with $x < x'$, then $K < x < x' \leq M$ and $x \mid x'$. So directed paths $x \rightarrow y \rightarrow x'$ and $x' \rightarrow y' \rightarrow x$ exist, and x, x' lie in the same SCC. Lying in the same SCC is an equivalence relation, so it propagates along paths of $G_{M,K}$. $\square$

Uses: independence of distinct edges; union bound.

Referee check. (1) Independence is used only for the events $E_y, y \in \mathcal{Y}$, for a fixed pair (x, x') ; these involve pairwise disjoint sets of edges. Across different pairs no independence is used, only the union bound. (2) The bound $|\mathcal{Y}| \geq N^{2\delta} - 2$ holds for every $x' \leq M$, so the union bound is uniform. (3) The vertices y may lie above M ; this is harmless, since paths may use any vertex of $V_{N,K}$. (4) If $K = 0$, the vertex 1 is included and the argument is unchanged.

4.2. Doubling.

Lemma 4.2 (Doubling). [PROVEN; new in this form, elementary] *Assume $M \geq 4(K+1)$, so that $Y \geq 2$. Then $S_M \neq \emptyset$, and S_M is contained in a single connected component of $G_{M,K}$. This component contains the unique power of 2 in $(M/2, M]$.*

Proof. Let 2^b be the power of 2 with $M/2 < 2^b \leq M$. Then $2^b > M/2 \geq 2(K+1) > K$ and $P^+(2^b) = 2 \leq Y$, so $2^b \in S_M$. For $x \in S_M$ let $\Omega'(x)$ be the number of odd prime factors of x counted with multiplicity. We show by induction on $\Omega'(x)$ that x is joined to 2^b by a path in $G_{M,K}$ that stays inside S_M .

Let $a \geq 0$ be maximal with $2^a x \leq M$, and put $y := 2^a x$. Then $M/2 < y \leq M$. The vertices $x, 2x, \dots, 2^a x = y$ lie in $(K, M]$, each divides the next, and they all lie in S_M (their odd part is that of x , and $2 \leq Y$). So they form a path in $G_{M,K}$. If $\Omega'(x) = 0$, then y is a power of 2 in $(M/2, M]$, so $y = 2^b$. Otherwise let p be an odd prime dividing x , and put $x_1 := y/p$. Then $x_1 \mid y$, $x_1 < y \leq M$, and

$$x_1 = \frac{y}{p} > \frac{M/2}{Y} = K+1,$$

using $p \leq P^+(x) \leq Y$. So $\{x_1, y\}$ is an edge of $G_{M,K}$. Moreover $P^+(x_1) \leq \max(P^+(x), 2) \leq Y$, so $x_1 \in S_M$, and $\Omega'(x_1) = \Omega'(x) - 1$. By induction x_1 is joined to 2^b , hence so is x . $\square$

Uses: nothing.

Corollary 4.3. [PROVEN] *Assume $M \geq 4(K+1)$. On $\mathcal{R}$ the set S_M lies in a single SCC of $\mathcal{D}_\rho(N, K)$, which we denote Σ . (On $\mathcal{R}^c$ put $\Sigma := \emptyset$.) We have $\Sigma \subset F_1$.*

Proof. The first claim follows from Lemmas 4.1 and 4.2. For $x \in S_M$ we have $P^+(x) \leq Y < N/(K+1)$, so $P^+(x) \leq B$ and $S_M \subset F_1 = V_1$. By Proposition 3.2, Σ lies in a single fibre, which must be V_1 . $\square$

4.3. Attachment. For $m \in V_{N,K}$ with $m > M$ let $D(m) := \{d \in S_M : d \mid m\}$ and $s(m) := |D(m)|$. Let A_m be the event that there are $d, d' \in D(m)$ with $m \rightarrow d$ and $d' \rightarrow m$.

Lemma 4.4 (Attachment). [PROVEN] *Assume $M \geq 4(K+1)$ and let $m \in V_{N,K}$ with $m > M$. If $s(m) \geq 1$, then $\mathbb{P}(A_m) = 1 - \rho^{s(m)} - (1 - \rho)^{s(m)} \geq 1 - 2\lambda^{s(m)}$. On $A_m \cap \mathcal{R}$ we have $m \in \Sigma$.*

Proof. Every $d \in D(m)$ satisfies $d \leq M < m$, so the edges $\{d, m\}$, $d \in D(m)$, are $s(m)$ distinct edges of $G_{N,K}$ with independent orientations. The complement A_m^c is the disjoint union of the events “all point $m \rightarrow d$ ” and “all point $d \rightarrow m$ ”, which have probabilities $(1 - \rho)^{s(m)}$ and $\rho^{s(m)}$. On $A_m \cap \mathcal{R}$ we have $m \rightarrow d$, a directed path $d \rightsquigarrow d'$ inside the strongly connected set $\Sigma \ni d, d'$, and $d' \rightarrow m$. This is a closed walk through m and d , so m lies in the SCC of d , which is Σ . $\square$

Uses: Corollary 4.3; independence of distinct edges.

Referee check. The events A_m and $\mathcal{R}$ are *not* independent: the edges $\{d, m\}$ may be among those used in the definition of $\mathcal{R}$. We never use independence between them. Below we only use $\mathbb{P}(A_m \cap \mathcal{R}) \geq \mathbb{P}(A_m) - \mathbb{P}(\mathcal{R}^c)$.

4.4. The arithmetic core.

Lemma 4.5 (Many divisors in the lower layer). [PROVEN; new] *Let $\eta \in (0, 1)$, $s_0 \in \mathbb{N}$ and $\delta \in (0, \eta/8]$, and put $z := N^{\delta/s_0}$. Assume*

$$(H1) \ K \leq N^{1-\eta}, \quad (H2) \ N^{\eta/4} \geq 4, \quad (H3) \ z \geq 2.$$

Let

$$\mathcal{U} := \{m \in \mathbb{Z} : N^{1-\delta} < m \leq N, \ P^+(m) \leq Y, \ \omega_z(m) \geq s_0\}.$$

Then:

- (a) $M \geq 4(K+1)$;
- (b) $\mathcal{U} \subset F_1$, and every $m \in \mathcal{U}$ satisfies $m > M$;
- (c) $s(m) \geq s_0 + 1$ for every $m \in \mathcal{U}$;
- (d) $|F_1 \setminus \mathcal{U}| \leq N^{1-\delta} + N \left(\frac{4\delta}{\eta} + \frac{2(1+C_1)}{\eta \log N} \right) + \#\{m \leq N : \omega_z(m) < s_0\}$.

Proof. Note first that $K+1 \leq 2N^{1-\eta}$ by (H1), and that $\eta - 2\delta \geq \frac{3}{4}\eta$ and $\eta - 6\delta \geq \frac{1}{4}\eta$ because $\delta \leq \eta/8$.

(a) $M/(K+1) \geq N^{1-2\delta}/(2N^{1-\eta}) = \frac{1}{2}N^{\eta-2\delta} \geq \frac{1}{2}(N^{\eta/4})^3 \geq 32$, by (H2).

(b) Let $m \in \mathcal{U}$. Then $m > N^{1-\delta} > N^{1-2\delta} = M$ and $m > N^{1-\delta} \geq N^{1-\eta} \geq K$, so $m \in V_{N,K}$. Also $P^+(m) \leq Y < N/(K+1)$, so $P^+(m) \leq B$ and $m \in F_1$.

(c) Fix $m \in \mathcal{U}$ and put $E_0 := N^{2\delta}$ and $E_1 := N^{1-\delta}/(K+1)$.

Step 1: divisors in $[E_0, E_1]$ give elements of $D(m)$. Let $e \mid m$ with $E_0 \leq e \leq E_1$ and put $d := m/e$. Then $d \leq N/E_0 = M$ and $d \geq m/E_1 > N^{1-\delta}(K+1)/N^{1-\delta} = K+1$. Also $P^+(d) \leq P^+(m) \leq Y$. Hence $d \in S_M$ and $d \mid m$, so $d \in D(m)$. Distinct e give distinct d .

Step 2: a reserve of small primes. Since $\omega_z(m) \geq s_0$, we may choose distinct primes $p_1, \dots, p_{s_0} \leq z$ dividing m . Put $r := p_1 \cdots p_{s_0}$, so $r \mid m$ and $r \leq z^{s_0} = N^\delta$. Then $m' := m/r$ is an integer with $m' \geq m/N^\delta > N^{1-2\delta}$.

Step 3: a base divisor $e_0 \mid m'$ with $E_0 \leq e_0 \leq E_1/z$. First note that

$$\frac{E_1}{z} = \frac{N^{1-\delta-\delta/s_0}}{K+1} \geq \frac{N^{1-2\delta}}{K+1} = 2Y.$$

Case (i): m' has a prime factor $q \geq E_0$. Then $E_0 \leq q \leq P^+(m) \leq Y \leq E_1/z$, and we take $e_0 := q$.

Case (ii): every prime factor of m' is $< E_0$. Write $m' = q_1 q_2 \cdots q_t$ with primes q_i (with multiplicity, in any order), and let $\pi_i := q_1 \cdots q_i$. Then $\pi_0 = 1 < E_0 \leq N^{1-2\delta} < m' = \pi_t$, where $E_0 \leq N^{1-2\delta}$ because $\delta \leq \frac{1}{4}$. Let j be the least index with $\pi_j \geq E_0$. Then $j \geq 1$ and $\pi_j = \pi_{j-1} q_j < E_0^2 = N^{4\delta}$. Moreover

$$\frac{E_1}{z} \geq \frac{N^{1-2\delta}}{K+1} \geq \frac{1}{2}N^{\eta-2\delta} = N^{4\delta} \cdot \frac{1}{2}N^{\eta-6\delta} \geq N^{4\delta} \cdot \frac{1}{2}N^{\eta/4} \geq N^{4\delta},$$

by (H2). So $e_0 := \pi_j$ satisfies $E_0 \leq e_0 \leq E_1/z$.

Step 4. Since $e_0 \mid m'$, we have $e_0 r \mid m$, so $e_0 p_i \mid m$ for each i . The s_0+1 numbers $e_0, e_0 p_1, \dots, e_0 p_{s_0}$ are distinct divisors of m , and they all lie in $[E_0, E_1]$ because $e_0 p_i \leq e_0 z \leq E_1$. By Step 1, $s(m) \geq s_0 + 1$.

(d) If $m \in F_1 \setminus \mathcal{U}$, then $m \leq N^{1-\delta}$, or $Y < P^+(m) \leq B$, or $\omega_z(m) < s_0$. The first set has at most $N^{1-\delta}$ elements. The second is empty if $Y \geq B$. Otherwise its size is at most $\sum_{Y < p \leq B} \lfloor N/p \rfloor \leq N \sum_{Y < p \leq B} \frac{1}{p}$. We have $Y \geq 2$ by (a). Moreover

$$\log B - \log Y \leq \log \frac{N}{K+1} - \log \frac{N^{1-2\delta}}{2(K+1)} = 2\delta \log N + \log 2,$$

and

$$\log Y \geq \log \frac{N^{1-2\delta}}{4N^{1-\eta}} = (\eta - 2\delta) \log N - \log 4 \geq \frac{3}{4}\eta \log N - \log 4 \geq \frac{1}{2}\eta \log N,$$

by (H2). By Lemma 2.1(b) and $\log(1+t) \leq t$,

$$\sum_{Y < p \leq B} \frac{1}{p} \leq \frac{\log B - \log Y}{\log Y} + \frac{C_1}{\log Y} \leq \frac{2\delta \log N + 1 + C_1}{\frac{1}{2}\eta \log N} = \frac{4\delta}{\eta} + \frac{2(1+C_1)}{\eta \log N}. \quad \square$$

Uses: Lemma 2.1(b).

Referee check. (1) No “almost all” statement is used inside the proof. Every $m \in \mathcal{U}$ has $s(m) \geq s_0+1$ deterministically, and the exceptional set is counted explicitly in (d). (2) The divisor e_0 is found by

a deterministic case analysis. Case (i) covers a prime factor of any size in $[E_0, P^+(m)]$, including sizes close to Y ; this is where the reserve primes must be tiny ($\leq z$), because e_0 may be as large as Y . (3) The bounds are uniform in $K \leq N^{1-\eta}$; K enters only through (H1).

4.5. The lower bound.

Proposition 4.6 (Lower bound). [PROVEN] *Assume the hypotheses of Lemma 4.5, and put*

$$\text{Err} := N^{1-\delta} + N \left(\frac{4\delta}{\eta} + \frac{2(1+C_1)}{\eta \log N} \right) + \#\{m \leq N : \omega_z(m) < s_0\} + 2\lambda^{s_0+1}N.$$

Then

$$(2) \quad \mathbb{E}[\Phi_K] \geq \Psi(N, B) - K - \text{Err} - N\mathbb{P}(\mathcal{R}^c),$$

$$(3) \quad \mathbb{E}[|F_1 \setminus \Sigma| \mathbf{1}_{\mathcal{R}}] \leq \text{Err}.$$

Here $\mathbb{P}(\mathcal{R}^c) \leq N^2 \exp(-q(N^{2\delta} - 2))$ by Lemma 4.1.

Proof. By Lemma 4.5(a), Corollary 4.3 and Lemma 4.4 apply. Let $X := \mathbf{1}_{\mathcal{R}} \cdot \#\{m \in \mathcal{U} : A_m \text{ occurs}\}$. On $\mathcal{R}$, every $m \in \mathcal{U}$ for which A_m occurs lies in Σ (Lemmas 4.5(b) and 4.4), so $X \leq |\Sigma| \leq \Phi_K$. On $\mathcal{R}^c$, $X = 0 \leq \Phi_K$. By Lemmas 4.5(c) and 4.4,

$$\mathbb{E}[\Phi_K] \geq \mathbb{E}[X] = \sum_{m \in \mathcal{U}} \mathbb{P}(A_m \cap \mathcal{R}) \geq \sum_{m \in \mathcal{U}} (\mathbb{P}(A_m) - \mathbb{P}(\mathcal{R}^c)) \geq |\mathcal{U}|(1 - 2\lambda^{s_0+1}) - N\mathbb{P}(\mathcal{R}^c).$$

Moreover $|\mathcal{U}| \geq |F_1| - |F_1 \setminus \mathcal{U}|$ and $|F_1| = \Psi(N, B) - \Psi(K, B) \geq \Psi(N, B) - K$. Also $2\lambda^{s_0+1}|\mathcal{U}| \leq 2\lambda^{s_0+1}N$. Combining these with Lemma 4.5(d) proves (2). For (3): on $\mathcal{R}$, $F_1 \setminus \Sigma \subset (F_1 \setminus \mathcal{U}) \cup \{m \in \mathcal{U} : A_m^c\}$. So

$$\mathbb{E}[|F_1 \setminus \Sigma| \mathbf{1}_{\mathcal{R}}] \leq |F_1 \setminus \mathcal{U}| + \sum_{m \in \mathcal{U}} \mathbb{P}(A_m^c) \leq |F_1 \setminus \mathcal{U}| + 2\lambda^{s_0+1}N \leq \text{Err}. \quad \square$$

Uses: Lemmas 4.1, 4.2, 4.4, 4.5; Corollary 4.3.

Referee check. (1) The only probabilistic inputs are exact computations for independent edges (Lemmas 4.1 and 4.4), the union bound, and $\mathbb{P}(A \cap \mathcal{R}) \geq \mathbb{P}(A) - \mathbb{P}(\mathcal{R}^c)$. (2) The inequality $\Phi_K \geq X$ holds pointwise on the whole probability space. (3) Proposition 4.6 is a statement for each fixed N , with no limits, so there is no interchange of limits. Uniformity in $K \leq N^{1-\eta}$ is explicit. Note that the term $\#\{m \leq N : \omega_z(m) < s_0\}$ does not depend on K .

5. PROOFS OF THE MAIN RESULTS

Proof of Theorem 1.3. The upper bound $\Phi_K \leq \Psi(N, B)$ is Proposition 3.2. For the lower bound, let N be large and put

$$\ell := \log \log N, \quad \eta := \frac{1}{\ell}, \quad \delta := \frac{1}{8\ell^2}, \quad s_0 := \max\left(1, \left\lceil \frac{\log \ell}{\log(1/\lambda)} \right\rceil\right), \quad z := N^{\delta/s_0}.$$

Then $\delta \leq \eta/8$ and $\lambda^{s_0} \leq 1/\ell$. Since $s_0 = O_\rho(\log \ell)$, there is $N_0(\rho)$ such that for $N \geq N_0(\rho)$ all of the following hold:

- (R1) $N^{\eta/4} = \exp(\frac{\log N}{4\ell}) \geq 4$ and $z = \exp(\frac{\log N}{8\ell^2 s_0}) \geq 2$;
- (R2) $L := \sum_{p \leq z} \frac{1}{p} \geq \log \log z - C_1 = \ell - \log(8\ell^2 s_0) - C_1 \geq \max(2s_0, \ell/2)$ (Lemma 2.1(c));
- (R3) $N^{1-\eta} \leq N/\ell$ and $N^{1-\delta} \leq N/\ell$;
- (R4) $2(1+C_1)/(\eta \log N) = 2(1+C_1)\ell/\log N \leq 1/(2\ell)$;
- (R5) $N \cdot N^2 \exp(-q(N^{2\delta} - 2)) \leq 1$, which holds because $N^{2\delta} = \exp(\log N/(4\ell^2))$ grows faster than any power of $\log N$;
- (R6) $\sqrt{N} + 2c_0 N/\log N \leq N/\ell$.

Case $K \leq N^{1-\eta}$. Hypotheses (H1)–(H3) of Lemma 4.5 hold by (R1). By Lemma 2.4 and (R2), $\#\{m \leq N : \omega_z(m) < s_0\} \leq 12N/L \leq 24N/\ell$. Also $4\delta/\eta = 1/(2\ell)$ and $2\lambda^{s_0+1} \leq 2/\ell$. Proposition 4.6 together with (R3)–(R5) gives

$$\Psi(N, B) - \mathbb{E}[\Phi_K] \leq \frac{N}{\ell} + \frac{N}{\ell} + N\left(\frac{1}{2\ell} + \frac{1}{2\ell}\right) + \frac{24N}{\ell} + \frac{2N}{\ell} + 1 \leq \frac{30N}{\ell}.$$

Case $K > N^{1-\eta}$. Then $B \leq N/(K+1) < N^\eta$. If $B < 2$, then $\Psi(N, B) = 1$. Otherwise Lemma 2.3 and (R6) give $\Psi(N, B) \leq \sqrt{N} + 2N(\eta \log N + c_0)/\log N \leq 3N/\ell$. In either case $0 \leq \Psi(N, B) - \mathbb{E}[\Phi_K] \leq \Psi(N, B) \leq 30N/\ell$. $\square$

Uses: Propositions 3.2, 4.6; Lemmas 2.1, 2.3, 2.4, 4.5.

Referee check. (1) The parameters η, δ, s_0, z depend on N only, not on K . So the two cases cover all K with one common bound. (2) The threshold $N_0(\rho)$ depends on ρ through s_0 (via λ) and through q in (R5). (3) Proposition 4.6 is applied at each fixed N , so no limit is interchanged.

Proof of Theorem 1.2. The deterministic bound is Proposition 3.2. The limit follows from Theorem 1.3, since $30/\log \log N \rightarrow 0$. $\square$

Remark 5.1 (Explicit form for $K \leq \sqrt{N} - 1$). [PROVEN] For $N \geq N'_0(\rho)$ and $0 \leq K \leq \sqrt{N} - 1$,

$$\left|1 - \frac{\mathbb{E}[\Phi_K]}{N} - \log \frac{\log N}{\log B}\right| \leq \frac{31}{\log \log N}.$$

Proof. Since $N/(K+1) \geq \sqrt{N}$, we have $B \geq \lfloor \sqrt{N} \rfloor$, and every prime $p > B$ satisfies $p > \sqrt{N}$. So each $n \leq N$ has at most one prime factor $> B$, and that factor divides n exactly once. Hence $N - \Psi(N, B) = \sum_{B < p \leq N} \lfloor N/p \rfloor$, which differs from $N \sum_{B < p \leq N} \frac{1}{p}$ by at most $\pi(N)$. By Lemma 2.1(b),(d) and $\log B \geq \frac{1}{4} \log N$ (for $N \geq 16$),

$$\left|1 - \frac{\Psi(N, B)}{N} - \log \frac{\log N}{\log B}\right| \leq \frac{C_1}{\log B} + \frac{\pi(N)}{N} \leq \frac{5C_1}{\log N} \leq \frac{1}{\log \log N}$$

for large N . Combine this with $0 \leq (\Psi(N, B) - \mathbb{E}[\Phi_K])/N \leq 30/\log \log N$ (Theorem 1.3). $\square$

Proof of Corollary 1.4. By Theorem 1.2, $\mathbb{E}[\Phi_K]/N = \Psi(N, y)/N + o(1)$ with $y := N/(K+1) \geq 1$.

If $\theta < 1$, put $u_N := \log N / \log y = (1 - \log(K+1)/\log N)^{-1}$. Then $u_N \geq 1$ and $u_N \rightarrow u := 1/(1-\theta)$. For large N we have $u_N \in [1, u+1]$, and Lemma 2.2 with $U = u+1$ gives $\Psi(N, N^{1/u_N})/N = \rho_{\text{Dick}}(u_N) + o(1) \rightarrow \rho_{\text{Dick}}(u)$, by continuity of ρ_{Dick} .

If $\theta = 1$, then $\log y / \log N \rightarrow 0$. Either $y < 2$ and $\Psi(N, y) = 1$, or Lemma 2.3 gives $\Psi(N, y)/N \leq N^{-1/2} + 2(\log y + c_0)/\log N \rightarrow 0$.

The profile is continuous and strictly decreasing on $[0, 1)$ by Lemma 2.2, since $\theta \mapsto 1/(1-\theta)$ is continuous and strictly increasing. At $\theta = 1$ it is continuous because $\rho_{\text{Dick}}(u) \leq 2/u \rightarrow 0$ (Lemma 2.3). For $\theta \leq \frac{1}{2}$ we have $u \in [1, 2]$, where $\rho_{\text{Dick}}(u) = 1 - \log u$. $\square$

Uses: Theorem 1.2; Lemmas 2.2, 2.3.

Proof of Corollary 1.5. Use the parameters and $N_0(\rho)$ from the proof of Theorem 1.3.

(a) By Proposition 3.2, $|\Phi_K - \Psi(N, B)| = \Psi(N, B) - \Phi_K$. Now apply Theorem 1.3 and Markov's inequality.

(b), (c): *Case $K \leq N^{1-\eta}$.* Work on $\mathcal{R}$, where $\Sigma \subset F_1$ is an SCC containing S_M (Corollary 4.3). Every SCC $C \neq \Sigma$ is disjoint from Σ and lies in a fibre V_n (Proposition 3.2). If $n = 1$, then $C \subset F_1 \setminus \Sigma$. If $n > 1$, then $|C| \leq K$ by Lemma 3.1(c). Hence every SCC other than Σ has at most $W + K$ vertices, where $W := |F_1 \setminus \Sigma|$.

- If $\Phi_K^* = \Sigma$, then $F_1 \triangle \Phi_K^* = F_1 \setminus \Sigma$, and $\Phi_K^{(2)} \leq W + K$.

- If $\Phi_K^* \neq \Sigma$ and $\Phi_K^* \subset V_1 = F_1$, then $|F_1 \triangle \Phi_K^*| = |F_1| - |\Phi_K^*| \leq |F_1| - |\Sigma| = W$. Also $\Phi_K^{(2)} \leq |\Phi_K^*| \leq W + K$.
- If $\Phi_K^* \subset V_n$ with $n > 1$, then $|\Sigma| \leq |\Phi_K^*| \leq K$. So $|F_1 \triangle \Phi_K^*| = |F_1| + |\Phi_K^*| \leq |\Sigma| + W + K \leq W + 2K$, and $\Phi_K^{(2)} \leq |\Phi_K^*| \leq K$.

Thus, on $\mathcal{R}$, $|F_1 \triangle \Phi_K^*| \leq W + 2K$ and $\Phi_K^{(2)} \leq W + K$. On $\mathcal{R}^c$ both quantities are at most $2N$. By (3) and the estimates in the proof of Theorem 1.3, $\mathbb{E}[W\mathbf{1}_{\mathcal{R}}] \leq \text{Err} \leq 28N/\ell$, and by (R5) $2N\mathbb{P}(\mathcal{R}^c) \leq 2 \leq N/\ell$. Hence $\mathbb{E}[|F_1 \triangle \Phi_K^*|] \leq 2K + 29N/\ell$ and $\mathbb{E}[\Phi_K^{(2)}] \leq K + 29N/\ell$ in this case.

Case $K > N^{1-\eta}$. Here $|F_1 \triangle \Phi_K^*| \leq |F_1| + \Phi_K \leq 2\Psi(N, B) \leq 6N/\ell$ and $\Phi_K^{(2)} \leq \Phi_K \leq \Psi(N, B) \leq 3N/\ell$. $\square$

Uses: Propositions 3.2, 4.6; Corollary 4.3; Lemma 3.1; Theorem 1.3.

Remark 5.2 (The term K in Corollary 1.5(c) is needed). [PROVEN] Suppose $K \rightarrow \infty$ and $K \leq \sqrt{N}/2$. Then $\mathbb{E}[\Phi_K^{(2)}] \geq (\frac{1}{2} - o(1))K$.

Proof. Let $N \geq 25$. Then $B \geq \frac{N}{K+1} - 1 \geq \frac{N}{\sqrt{N}/2+1} - 1 \geq \sqrt{N} + 1$. Let p be a prime in $(B, 2B]$, which exists by Bertrand's postulate. For $1 \leq s \leq N/p$ we have $s < N/B < \sqrt{N} < B < p$. So s is B -smooth and $p \nmid s$, which gives $R_B(ps) = p$. Also $ps \geq p > K$. Together with Lemma 3.1(c) this shows $V_p = \{ps : 1 \leq s \leq N'\}$ with $N' := \lfloor N/p \rfloor$. Since $ps \mid ps'$ if and only if $s \mid s'$, the map $s \mapsto ps$ is an isomorphism from $G_{N'}$ onto $G_N[V_p]$ that preserves the default orientation. Hence the restriction of $\mathcal{D}_\rho(N, K)$ to V_p has the law of $\mathcal{D}_\rho(N')$. As V_p is a union of connected components of $G_{N,K}$, its SCCs are SCCs of $\mathcal{D}_\rho(N, K)$. We have $N' \geq \lfloor N/(2B) \rfloor \geq \lfloor (K+1)/2 \rfloor \rightarrow \infty$. So by [9, Corollary 2] the largest SCC inside V_p has expected size $(1 - o(1))N'$.

Next, $\Psi(N, B) \geq \Psi(N, \sqrt{N}) \geq N - N \sum_{\sqrt{N} < p \leq N} \frac{1}{p} \geq N/4$ for large N , by Lemma 2.1(b). By Theorem 1.2 and Markov's inequality,

$$\mathbb{P}(\Phi_K \leq K) \leq \frac{\mathbb{E}[\Psi(N, B) - \Phi_K]}{N/4 - K} \rightarrow 0.$$

On the event $\{\Phi_K > K\}$ the largest SCC lies in $F_1 = V_1$ (Lemma 3.1(c)), which is disjoint from V_p , so $\Phi_K^{(2)} \geq \#\Phi(\mathcal{D}_\rho(N, K)[V_p])$. Therefore

$$\mathbb{E}[\Phi_K^{(2)}] \geq \mathbb{E}[\#\Phi(\mathcal{D}_\rho(N, K)[V_p])] - N' \mathbb{P}(\Phi_K \leq K) = (1 - o(1))N' \geq (\frac{1}{2} - o(1))K. \quad \square$$

Remark 5.3 (Uniformity in ρ ; slowly vanishing ρ). [PROVEN] The proof depends on ρ only through $q = \rho(1 - \rho)$ and $\lambda = \max(\rho, 1 - \rho)$. So Theorems 1.2 and 1.3 hold uniformly for $\rho \in [\rho_0, 1 - \rho_0]$, with $N_0 = N_0(\rho_0)$. More generally, Theorem 1.2 remains true for $\rho = \rho_N$ provided $\min(\rho_N, 1 - \rho_N) \log \log N \rightarrow \infty$. To see this, apply Proposition 4.6 with fixed η and $\delta \leq \eta/8$ and with $s_0 := \lfloor \frac{1}{4} \log \log N \rfloor$. Then Lemma 2.1(c) gives $L \geq \log \log N - \log(s_0/\delta) - C_1 \geq 2s_0$ for large N , so the ω_z term is $\leq 12N/L = o(N)$. Next, $2\lambda^{s_0+1} \leq 2 \exp(-\min(\rho_N, 1 - \rho_N)s_0) \rightarrow 0$. Finally $N\mathbb{P}(\mathcal{R}^c) \leq N^3 \exp(-q(N^{2\delta} - 2)) \rightarrow 0$, since $q \geq \frac{1}{2} \min(\rho_N, 1 - \rho_N) \geq 1/\log \log N$ for large N . Letting $\delta \rightarrow 0$ and then $\eta \rightarrow 0$ gives the claim, with the case $K > N^{1-\eta}$ handled as before.

6. REMOVING THE VERTICES OF HIGHEST DEGREE

For $T \subset [N]$ write $\deg_T(v)$ for the degree of v in $G_N - T$. In G_N ,

$$\deg(m) = (\tau(m) - 1) + (\lfloor N/m \rfloor - 1).$$

Put $\tau_N := \max_{n \leq N} \tau(n)$. Then $\deg_T(m) \leq \deg(m) \leq \tau_N + N/m$ for every T and $m \notin T$.

Lemma 6.1 (Monotonicity). [PROVEN] If D is a directed graph and $U \subset V(D)$, then $\#\Phi(D[U]) \leq \#\Phi(D)$.

Proof. An SCC of $D[U]$ is strongly connected in D , so it is contained in an SCC of D . $\square$

Lemma 6.2 (Sandwich). [PROVEN] *Let $0 < \varepsilon \leq \frac{1}{2}$ and $1 \leq K < N$ with $\frac{N}{K} \cdot \frac{\varepsilon}{2} \geq \tau_N + 5$. Let T be a static or an adaptive top- K set as in Corollary 1.6. Then*

$$\{1, \dots, \lfloor (1 - \varepsilon)K \rfloor\} \subset T \subset \{1, \dots, \lfloor (1 + \varepsilon)K \rfloor\}.$$

Proof. Static, upper inclusion. Suppose $w \in T$ with $w > (1 + \varepsilon)K$. Then $|T \cap [1, K]| \leq K - 1$, so some $v \leq K$ has $v \notin T$. Now

$$\deg(v) \geq \lfloor N/v \rfloor - 1 \geq N/K - 2 \quad \text{and} \quad \deg(w) < \tau_N + N/((1 + \varepsilon)K).$$

The static condition gives $\deg(w) \geq \deg(v)$, hence $\frac{N}{K} \cdot \frac{\varepsilon}{1 + \varepsilon} < \tau_N + 2$. This contradicts the hypothesis, since $\frac{\varepsilon}{1 + \varepsilon} \geq \frac{\varepsilon}{2}$.

Static, lower inclusion. Suppose $v \leq (1 - \varepsilon)K$ and $v \notin T$. Every $t \in T$ satisfies $\tau_N + N/t \geq \deg(t) \geq \deg(v) \geq N/v - 2 \geq \frac{N}{(1 - \varepsilon)K} - 2$. If $t > (1 - \frac{\varepsilon}{2})K$, this gives $\frac{N}{K} \left(\frac{1}{1 - \varepsilon} - \frac{1}{1 - \varepsilon/2} \right) < \tau_N + 2$. The left side is at least $\frac{N}{K} \cdot \frac{\varepsilon}{2}$, a contradiction. So $T \subset [1, (1 - \frac{\varepsilon}{2})K]$. But this set has fewer than K elements, which contradicts $|T| = K$.

Adaptive, upper inclusion. We show by induction on $j < K$ that $T_j := \{u_1, \dots, u_j\} \subset [1, (1 + \varepsilon)K]$ implies $u_{j+1} \leq (1 + \varepsilon)K$. Since $|T_j| < K$, some $v \leq K$ has $v \notin T_j$. No multiple of v in $((1 + \varepsilon)K, N]$ has been removed, so

$$\deg_{T_j}(v) \geq \frac{N - (1 + \varepsilon)K}{v} - 1 \geq \frac{N}{K} - 3.$$

Every $w > (1 + \varepsilon)K$ has $\deg_{T_j}(w) < \tau_N + N/((1 + \varepsilon)K)$, and this is at most $N/K - 3$ by the hypothesis. Hence the maximiser u_{j+1} satisfies $u_{j+1} \leq (1 + \varepsilon)K$.

Adaptive, lower inclusion. Suppose $v \leq (1 - \varepsilon)K$ and $v \notin T_K$. For every $j < K$, using $T_j \subset [1, (1 + \varepsilon)K]$,

$$\deg_{T_j}(v) \geq \frac{N - (1 + \varepsilon)K}{v} - 1 \geq \frac{N - 2K}{(1 - \varepsilon)K} - 1.$$

Also $\tau_N + N/u_{j+1} \geq \deg(u_{j+1}) \geq \deg_{T_j}(u_{j+1}) \geq \deg_{T_j}(v)$. If $u_{j+1} > (1 - \frac{\varepsilon}{2})K$, then

$$\frac{N}{K} \left(\frac{1}{1 - \varepsilon} - \frac{1}{1 - \varepsilon/2} \right) < \tau_N + 1 + \frac{2}{1 - \varepsilon} \leq \tau_N + 5.$$

The left side is at least $\frac{N}{K} \cdot \frac{\varepsilon}{2}$, a contradiction. So all K vertices $u_1, \dots, u_K$ lie in $[1, (1 - \frac{\varepsilon}{2})K]$, which has fewer than K elements. This is impossible. $\square$

Uses: the degree formula.

Proof of Corollary 1.6. Let $\varepsilon = \frac{1}{4}$ and $K_{\pm} := \lfloor (1 \pm \varepsilon)K \rfloor$. If $K = 0$ there is nothing to prove. Since $\log(K + 1)/\log N \rightarrow \theta < 1$, we have $N/K = N^{1 - \theta + o(1)}$. By Lemma 2.6 with exponent $(1 - \theta)/2$, $\tau_N = O(N^{(1 - \theta)/2})$. So the hypothesis of Lemma 6.2 holds for large N , and $V_{N, K_+} \subset [N] \setminus T \subset V_{N, K_-}$. Since T does not depend on the orientation, Lemma 6.1 applied to the single graph $\mathcal{D}_{\rho}(N)$ gives $\Phi_{K_+} \leq \#\Phi(\mathcal{D}_{\rho}(N) - T) \leq \Phi_{K_-}$ pointwise. Since $\log(K_{\pm} + 1)/\log N \rightarrow \theta$, Corollary 1.4 applies to $K_{\pm}$ and gives the limit.

For the last claim, let $m \leq K < m'$ with $K \leq N^{1/2 - \epsilon}$. Then $\deg(m) - \deg(m') \geq \lfloor N/K \rfloor - \lfloor N/(K + 1) \rfloor - \tau_N \geq \frac{N}{K(K + 1)} - 1 - \tau_N > 0$ for large N , by Lemma 2.6 with exponent ϵ . So the K vertices of largest degree are exactly $1, \dots, K$. $\square$

Uses: Lemmas 2.6, 6.1, 6.2; Corollary 1.4.

Referee check. The attack set T must not depend on the orientation. Otherwise $\mathcal{D}_{\rho}(N) - T$ is not an independently oriented copy of $G_N - T$, and the coupling argument breaks down. This is why in- and out-degree attacks are excluded.

7. REMARKS AND OPEN PROBLEMS

Remark 7.1 (Edge share of the hubs). [PROVEN] An edge meets $\{1, \dots, K\}$ if and only if its smaller endpoint d is at most K . So $\{1, \dots, K\}$ meets exactly $\sum_{d \leq K} (\lfloor N/d \rfloor - 1) = N \log K + O(N)$ of the $\sum_{d \leq N} (\lfloor N/d \rfloor - 1) = N \log N + O(N)$ edges, for $1 \leq K \leq N$.

Remark 7.2 (Why a hub clique does not suffice for $\theta > 1/3$). [SKETCH; not used] A natural first approach takes the hub set $H = (K, N^{1/2-\delta}]$. It is strongly connected w.h.p., because any two hubs have at least $N^{2\delta}$ common multiples. One then attaches every m with many divisors in H . This captures only the integers having a divisor in $(K, N^{1/2-\delta}]$. For $\theta \leq 1/3$ these are asymptotically almost all of F_1 (compare the sub-sum lemma, Appendix A). For $1/3 < \theta < 1/2$, however, integers $m = p_1 p_2 p_3 s$ with primes $p_i \in (N^{1/4}, N^{1/3}]$ and a cofactor $s \leq N^\gamma$, $\gamma < \theta - \frac{1}{3}$, lie in F_1 and have no divisor in $(K, N^{1/2-\delta}]$. Their number is $\gg N$ by a routine Mertens-type triple sum, which we do not write out. The method above avoids this obstruction. Such an m either lies in S_M itself or attaches through its divisor $d = m/p_1 = p_2 p_3 s \in (K, M]$, which lies in S_M . The connectivity of the lower layer is supplied deterministically by Lemma 4.2, not by a clique.

Remark 7.3 (Relation to [9]). The lower bound of [9] uses the triangles $\{1, d, m\}$ and needs the vertex 1. The present argument uses 2-paths through common *multiples* of divisibility pairs in the lower layer. It gives a new proof of [9, Corollary 2], the case $K = 0$ of Corollary 1.4, and extends it to all K .

7.1. Open problems.

- (O1) [OPEN] *True rate*. Determine the order of $\Psi(N, N/(K+1)) - \mathbb{E}[\Phi_K]$. Theorem 1.3 gives $O(N/\log \log N)$. Vertices of degree 1 in F_1 are never in an SCC with at least two vertices. This gives $N - \mathbb{E}[\Phi_0] \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$ (the primes in $(N/2, N]$) and $\Psi(N, B) - \mathbb{E}[\Phi_1] \geq \pi(N/2) - \pi(N/3) \sim N/(6 \log N)$ (the primes in $(N/3, N/2]$). Is the error $N(\log N)^{-1+o(1)}$, uniformly in $K \leq N^{1-\eta}$? The bottleneck in our argument is the variance bound (Lemma 2.4) and the thin layer $Y < P^+(m) \leq B$.
- (O2) [OPEN] *Fluctuations*. Find the variance and the limiting distribution of Φ_K ; this is [9, Question 21] in the presence of hub removal. Corollary 1.5 gives concentration only at scale $o(N)$.
- (O3) [OPEN] *Orientation-dependent attacks*. Determine the limit profile when hubs are removed by in-degree or out-degree in the realised orientation, which is adaptive to the randomness.
- (O4) [OPEN] *Small ρ* . Remark 5.3 allows $\min(\rho, 1 - \rho) \gg 1/\log \log N$. Where does the profile change when $\rho = \rho_N \rightarrow 0$ faster?
- (O5) [OPEN] *Diameter*. What is the diameter of the giant SCC after hub removal? Compare [9, Conjecture 6].

APPENDIX A. THE SUB-SUM LEMMA

The preliminary notes suggested proving the case $\theta \leq 1/3$ through the Poisson–Dirichlet description of the large prime factors together with the following deterministic lemma. Our proof of Theorem 1.2 makes this route unnecessary, but we record the lemma with a proof.

Lemma A.1 (Sub-sum lemma). [PROVEN; elementary] *Let $(x_i)_{i \geq 1}$ be a finite or infinite sequence with $x_1 \geq x_2 \geq \dots > 0$ and $\sum_i x_i = 1$, and let $0 < \theta \leq 1/3$. If $x_1 < 1 - \theta$, then there is a set I of indices with $\sum_{i \in I} x_i \in [\theta, \frac{1}{2}]$. This fails for every $\theta \in (\frac{1}{3}, \frac{1}{2}]$: take $(x_1, x_2, x_3) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.*

Proof. We use that if $S = \sum_{i \in I} x_i \in [\theta, 1 - \theta]$, then either $S \leq \frac{1}{2}$, or $1 - S = \sum_{i \notin I} x_i \in [\theta, \frac{1}{2}]$.

If $x_1 \geq \theta$, take $I = \{1\}$; then $x_1 \in [\theta, 1 - \theta]$.

If $x_1 < \theta$, then all $x_i < \theta$. The partial sums $S_n = x_1 + \dots + x_n$ increase from $S_0 = 0$ to $1 > \theta$. Let n be minimal with $S_n \geq \theta$. Then $S_n = S_{n-1} + x_n < 2\theta \leq 1 - \theta$. So $S_n \in [\theta, 1 - \theta]$, and we take $I = \{1, \dots, n\}$ or its complement.

For the counterexample with $\theta \in (\frac{1}{3}, \frac{1}{2}]$: $x_1 = \frac{1}{3} < 1 - \theta$, and the sub-sums are $0, \frac{1}{3}, \frac{2}{3}, 1$, none of which lies in $[\theta, \frac{1}{2}]$. $\square$

APPENDIX B. STATUS OF ALL CLAIMS

Claim	Status	Notes / dependencies
Lemma 2.1 (Mertens, Chebyshev)	PROVEN (cited)	classical [11, 6, 12]
Lemma 2.2 (Dickman, uniform on $[1, U]$)	PROVEN (cited)	[4, 3, 7]; needed only to identify the limit
Lemma 2.3 ($\Psi \leq \sqrt{x} + 2x(\log y + c_0)/\log x$)	PROVEN	standard; uses Mertens (a)
Lemma 2.4 (variance of ω_z)	PROVEN	special case of Turán–Kubilius
Lemma 2.6 ($\tau(n) \ll_\varepsilon n^\varepsilon$)	PROVEN	standard
Lemma 3.1 (fibres)	PROVEN	essentially known (cf. [10])
Proposition 3.2 ($\Phi_K \leq \Psi(N, B)$, $\Phi_K \leq \max(F_1 , K)$)	PROVEN	deterministic
Corollary 3.3 (limsup)	PROVEN	
Remark 3.4 (equivalent forms)	PROVEN	Markov
Lemma 4.1 (robust edges)	PROVEN	new; union bound, no cross-pair independence
Lemma 4.2 (doubling)	PROVEN	new in this form; deterministic
Corollary 4.3	PROVEN	
Lemma 4.4 (attachment)	PROVEN	$\mathbb{P}(A \cap \mathcal{R}) \geq \mathbb{P}(A) - \mathbb{P}(\mathcal{R}^c)$
Lemma 4.5 (arithmetic core)	PROVEN	new; deterministic, explicit exceptional set
Proposition 4.6 (lower bound, explicit)	PROVEN	non-asymptotic
Theorem 1.3 (rate $30N/\log \log N$)	PROVEN	new
Theorem 1.2 (uniform asymptotic)	PROVEN	new; answers [9, Q. 20]
Remark 5.1 (explicit loss $\log \frac{\log N}{\log B} + O(1/\log \log N)$ for $K < \sqrt{N}$)	PROVEN	new; the rate part of T5
Corollary 1.4 (profile $\rho_{\text{Dick}}(1/(1-\theta))$, all $\theta \in [0, 1]$)	PROVEN	new; contains $\theta \leq 1/3$, $1/3 < \theta < 1/2$, $\theta \geq 1/2$
Corollary 1.5 (giant $\approx F_1$, second $\text{SCC} \leq K + o(N)$)	PROVEN	new
Remark 5.2 (term K needed)	PROVEN	uses [9, Cor. 2]
Remark 5.3 (uniform in ρ ; $\rho_N \gg 1/\log \log N$)	PROVEN	
Lemma 6.1, Lemma 6.2	PROVEN	
Corollary 1.6 (static/adaptive degree attacks)	PROVEN	degrees taken in G_N
Remark 7.1 (edge share θ)	PROVEN	
Remark 7.2 (hub clique loses for $\theta > 1/3$)	SKETCH	gap: the positive-density count of the $m = p_1 p_2 p_3 s$ is not written out; not used anywhere
Lemma A.1 (sub-sum lemma, sharp)	PROVEN	not needed for the main results

Claim	Status	Notes / dependencies
(O1)–(O5)	OPEN	

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